📄 Product Requirements Document

📋 Document Information

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🎯 Executive Summary

Problem
Small and mid-size SaaS companies struggle to provide scalable, cost-efficient customer support while maintaining control over their data and AI costs.

Solution
SupportBase is a customer support platform combining **AI-powered RAG chatbots** with a **human-in-the-loop ticketing system**, delivered via a **drop-in JavaScript SDK** and a **Bring Your Own AI (BYO-AI)** architecture.

Impact

- Reduce support ticket volume by **40–60%**
- Reduce first-response time by **80%**
- Enable AI support without vendor lock-in or token markup

📊 Problem Statement

Current State Analysis

- SMB SaaS teams rely on:
  - Expensive AI helpdesks with token markup
  - Manual support via email/Discord
  - Chatbots with no escalation to humans

- Existing solutions:
  - Lock users into proprietary AI models
  - Store customer data on vendor-controlled systems
  - Offer poor developer experience

Key Pain Points

1. **High AI Costs**
   AI support tools add 30–70% markup on tokens.

2. **No Data Ownership**
   Support data and embeddings are locked into vendors.

3. **Broken AI → Human Handoff**
   AI chatbots fail silently without proper escalation.

4. **Poor SDK Experience**
   Difficult to embed and customize for developers.

Root Cause Analysis

- Support tools are built for enterprises, not developers
- AI-first platforms optimize margins, not transparency
- Lack of modular architecture separating AI, storage, and UX

Cost of Inaction

- Increasing support headcount
- Slower response times
- Poor customer retention
- Unpredictable AI costs

🎯 Goals and Success Metrics

Primary Objective

Enable SMB SaaS teams to deploy **AI + human customer support** with **full data and cost control** in under **10 minutes**.

🥇 Level 0 Metrics (Primary KPIs)

- **Time to First Response**: 24 hrs → **<5 min** (3 months)
- **AI Resolution Rate**: 0% → **50%+** (3 months)

🥈 Level 1 Metrics (Secondary KPIs)

- **Support Ticket Volume**: −40% (3 months)
- **Cost per Conversation**: −60% vs traditional tools

🥉 Level 2 Metrics (Engagement & Quality)

- **Chat Widget Engagement Rate**: >25%
- **User Satisfaction (CSAT)**: >4 / 5

🛡️ Guardrail Metrics

- AI hallucination reports <5%
- Widget load time <200ms

👥 Target Audience

Primary Users

Indie Developers & SMB SaaS Founders

- Size: 1–20 employees
- Tech-savvy, cost-conscious
- Limited support resources
- Prefer self-serve tools

Current Behavior

- Use email, Discord, Intercom alternatives
- Avoid AI tools due to cost or complexity

Motivation

- Reduce support workload
- Improve customer experience
- Keep full control over AI & data

Secondary Users

End Customers of SaaS Products

- Users seeking fast, accurate support
- Expect instant answers with escalation

User Personas

Persona 1: Indie SaaS Founder
Goal: Reduce support load without hiring
Frustration: Expensive AI tools, vendor lock-in

Persona 2: Developer Advocate
Goal: Embed support quickly
Frustration: Poor SDKs and limited customization

💡 Solution Overview

High-Level Approach

- Drop-in JavaScript SDK for chat + ticketing
- RAG-based AI answers using user-owned models
- Human escalation when AI confidence is low
- Transparent, modular backend architecture

Key Principles

- BYO-AI (Bring Your Own API Keys)
- No token markup
- Developer-first UX
- Human override always available

Solution Components

- Chat Widget SDK
- RAG Retrieval Engine
- Vector Store (Per Project)
- Ticketing System
- Admin Dashboard
- OpenRouter LLM Gateway

🔧 Detailed Requirements

Feature 1: JavaScript Chat Widget SDK

Objective
Enable instant AI + human support embedding.

User Story
As a developer, I want to embed a support widget with one script so that I can offer AI and human support instantly.

Functional Requirements

- Single JS script/SDK installation
- Set Open Router parameters like cost, model, etc.
- Theme customization
- Streaming AI responses

Acceptance Criteria

- Given SDK installed, when user asks a question, then AI responds in real-time
- Given AI fails, when confidence is low, then ticket is created

Priority: High
Effort Estimate: 2 weeks

Feature 2: RAG-based AI Answering

Objective
Provide accurate answers using customer's own data.

Functional Requirements

- Document ingestion (Markdown, PDF, URLs)
- Chunking & embeddings
- Top-K retrieval
- Prompt injection protection

Acceptance Criteria

- Given relevant data exists, AI answers reference correct context
- Given no context, fallback to ticket creation

Priority: High
Effort Estimate: 2 weeks

Feature 3: Human-in-the-Loop Ticketing

Objective
Ensure unresolved queries reach humans.

Functional Requirements

- Automatic ticket creation
- Manual takeover from chat
- Internal notes
- Status tracking

Priority: High
Effort Estimate: 1.5 weeks

⚙️ Technical Requirements

Architecture Overview

- Multi-tenant backend
- Per-project vector stores

- OpenRouter LLM routing
- Stateless chat workers

Frontend Requirements

- <50KB SDK bundle
- No framework dependency
- Works on all modern browsers

Backend Requirements

- REST + WebSocket APIs
- Streaming responses
- Vector DB per project
- Secure key isolation

Third-Party Dependencies

- OpenRouter
- OpenAI / Anthropic (via user keys)
- Vector DB (Qdrant / Weaviate)

📅 Implementation Plan

Phase 1: MVP

Timeline: 8 weeks
Scope:

- Chat widget
- RAG pipeline
- Ticket creation
- Admin dashboard

Success Criteria

- First 10 live customers
- <5 min first response time

Phase 2: Enhancements

- Multi-language support
- Analytics dashboard
- SLA rules

Phase 3: Future Iterations

- Voice support
- CRM integrations
- Auto fine-tuning

🎯 Success Criteria

- AI resolves >50% queries
- Customers onboard in <10 minutes
- Monthly tickets reduced >30%

⚠️ Risks and Mitigation

Technical Risks

Risk: AI hallucinations
Mitigation: Strict RAG grounding + confidence thresholds

Business Risks

Risk: Competition from incumbents
Mitigation: Open, transparent pricing & dev-first UX

📈 Analytics and Measurement

Key Events

- Chat opened
- AI answered
- Ticket created
- Human reply sent

Dashboards

- AI resolution rate
- Cost per conversation
- Support backlog

🚀 Go-to-Market Strategy

Launch Strategy

- Indie Hacker launch
- GitHub README-driven onboarding
- Free tier with BYO-AI

Communication Plan

Internal: Weekly demos
External: Docs, blog posts, demos